

SCOPE-OPSD: FISHER-CONDITIONED PRIVILEGED SUBSPACES FOR ON-POLICY SELF-DISTILLATION

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ABSTRACT

On-policy self-distillation (OPSD) scores student-generated prefixes with a solution-conditioned self-teacher, yet transfers supervision only through next-token probabilities. We ask whether the aligned final-layer discrepancy offers a useful second channel, and how to test that channel without confusing its geometry with auxiliary strength. SCOPE-OPSD projects the privileged teacher–student residual onto a frozen rank-64 factor estimated from residual covariance and language-model-head Fisher sensitivity. It reuses the forwards already required by OPSD and adds neither rollouts nor inference-time modules. A matched Random control preserves the structured factor’s rank and nonzero spectrum and uses per-arm gradient-RMS calibration, isolating the effect of the data-dependent orientation. Across the complete 25/50/75/100-step trajectories for Qwen3-1.7B, 4B, and 8B, Structured is never below Pure OPSD, with strict gains in 11 of the 12 model–checkpoint combinations and an exact tie at 4B step 25. Structured also exceeds matched Random in 10 of the 12 combinations. At step 75 on Qwen3-1.7B, Structured exceeds matched Random by 1.39 Macro Avg@12 points in each of two independent training reruns. A cross-fitted diagnostic also shows 4.40× greater held-out privileged-gap capture than the matched random orientation. The results support a compact, Fisher-conditioned privileged subspace for short-budget OPSD.

1 INTRODUCTION

On-policy distillation trains a student on the states it actually visits (Agarwal et al., 2024; Song & Zheng, 2026). This is especially relevant to long-form reasoning, where a fixed teacher solution may look quite different from a trajectory sampled by the student. On-Policy Self-Distillation (OPSD) uses one model under two information conditions (Zhao et al., 2026a): the student sees only the problem, whereas a frozen self-teacher also receives a verified solution. At each student-generated prefix, the privileged teacher provides a dense next-token target over the full vocabulary. The method is simple and effective under a short post-training budget.

OPSD, however, uses privileged teacher information only through the token probabilities produced by the language-model head. The final hidden states used to produce those probabilities are already aligned by token and available in the same forward passes, but are otherwise discarded. Recent work has shown that hidden-space supervision can benefit on-policy distillation. OPRD aligns representations across layers (Yang et al., 2026a), and PHF augments OPSD with transition and trajectory geometry (Li et al., 2026b). A basic attribution question remains: does a hidden auxiliary

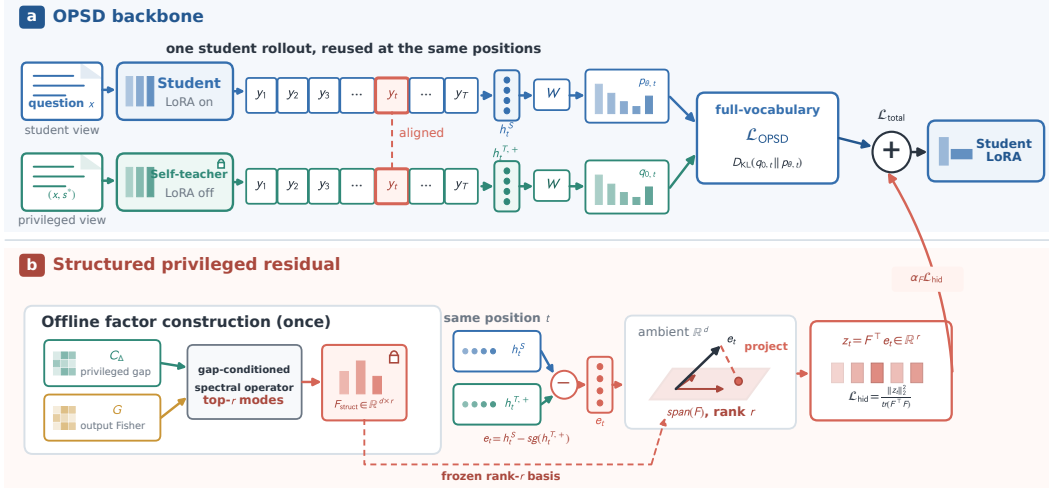


Figure 1: Overview of SCOPE-OPSD. The blue and green paths are the existing OPSD computation: one student rollout is rescored by the student and a frozen solution-conditioned self-teacher at the same token positions. The coral branch projects their final-state residual through a frozen rank- r factor F_{struct} and adds the projected loss to full-vocabulary OPSD. The factor is fitted once from privileged-gap covariance C_Δ and LM-head Fisher geometry G . Only the student LoRA is updated.

help because its orientation captures privileged information, or because any auxiliary gradient regularizes the student?

We study the narrowest hidden interface already present in OPSD: the final state immediately before the language-model head. At each rollout position, we subtract the stopped-gradient privileged-teacher state from the student state and project the residual through a fixed low-rank factor. The factor is estimated once from two calibration statistics. Privileged-gap covariance marks directions in which the two information conditions differ; output Fisher geometry measures how local hidden perturbations are expressed through the frozen language-model head. A smooth spectral filter combines them while suppressing both output-invisible directions and the most strongly expressed directions already emphasized by the output loss.

The control is as important as the auxiliary. We replace the structured factor with a random orientation that preserves its rank and nonzero spectrum, then calibrate each hidden-loss coefficient to the same initial gradient RMS. Structured versus Random therefore tests the learned orientation while holding the principal scale variables fixed. Pure OPSD provides the output-only reference.

We use step 75 as the common checkpoint across model scales. At this checkpoint, Structured improves matched Random by 1.39, 0.37, and 0.93 percentage points on Qwen3-1.7B, 4B, and 8B, respectively; the corresponding gains over Pure OPSD are 1.85, 1.67, and 0.83 points. We also report every saved checkpoint through step 100 rather than selecting a separate point for each benchmark. Finally, a cross-fitted offline test shows that the structured factor captures $4.40\times$ the held-out privileged-gap score of its matched random control.

Our contributions are:

- a final-layer privileged-residual objective that reuses OPSD’s aligned forwards and adds no rollout or inference-time component;
- a frozen Fisher-conditioned rank-64 subspace, paired with spectrum- and gradient-matched controls that isolate its orientation; and
- matched evidence across three Qwen3 scales, including complete checkpoint curves and component controls that place GapOnly between Random and Structured while favoring rank 64 over the tested ranks 32 and 128 and full-state matching.

2 BACKGROUND AND PROBLEM FORMULATION

Let x be a question, s a verified reference solution, and $y = (y_1, \dots, y_T)$ a trajectory sampled from the current student p_θ :

$$y \sim p_\theta(\cdot | x). \quad (1)$$

At position t , the student predicts from $(x, y_{<t})$. A frozen privileged self-teacher q_0 receives the same student prefix as well as s . Following classical distribution matching and its on-policy extension (Hinton et al., 2015; Agarwal et al., 2024), OPSD minimizes a token-averaged forward divergence (Zhao et al., 2026a),

$$\mathcal{L}_{\text{OPSD}} = \mathbb{E}_{(x,s),y} \left[\frac{1}{T} \sum_{t=1}^T D_{\text{KL}}(q_0(\cdot | x, s, y_{<t}) \| p_\theta(\cdot | x, y_{<t})) \right]. \quad (2)$$

Our implementation follows the original full-vocabulary objective and applies the same pointwise upper clip of 0.05 to individual vocabulary contributions. The teacher remains the step-zero base model with the trainable student adapter disabled.

Write $h_t^S, h_t^{T,+} \in \mathbb{R}^d$ for the final pre-head hidden states of the student and privileged teacher. With language-model head $W \in \mathbb{R}^{|\mathcal{V}| \times d}$, OPSD observes these states through Wh_t^S and $Wh_t^{T,+}$. This output projection is appropriate for next-token matching, but it leaves two questions unanswered. First, does the privileged residual

$$e_t = h_t^S - \text{sg}(h_t^{T,+}) \quad (3)$$

carry training information not already exploited by equation (2)? Second, if it does, which directions of e_t should be supervised?

The second question creates a confound. Hidden losses with identical scalar coefficients can have very different gradient magnitudes because their operators have different spectra and align differently with the model Jacobian. A fair orientation comparison therefore requires more than assigning the same coefficient to two projectors. We explicitly calibrate the initial auxiliary gradient and treat orientation quality as a separate hypothesis from the usefulness of hidden supervision itself.

3 FISHER-CONDITIONED PRIVILEGED SUPERVISION

3.1 A MINIMAL FINAL-LAYER AUXILIARY

Figure 1 summarizes the separation between the existing OPSD computation, the one-time factor construction, and the online objective.

For a fixed factor $F \in \mathbb{R}^{d \times r}$, we define

$$\mathcal{L}_{\text{hid}}(F) = \mathbb{E}_{(x,s),y} \left[\frac{1}{T} \sum_{t=1}^T \frac{\|F^\top e_t\|_2^2}{\text{tr}(F^\top F)} \right]. \quad (4)$$

The teacher state is stop-gradient. The trace normalization makes the reported loss less sensitive to a uniform rescaling of F ; it does not by itself match the gradient delivered to the trainable adapter. The training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{OPSD}} + \alpha_F \mathcal{L}_{\text{hid}}(F), \quad (5)$$

where α_F is calibrated for each control as described below.

The auxiliary uses the same token positions and forwards as OPSD. Its only new online operation is the matrix multiplication $F^\top e_t$; it requires no additional model forward, rollout, reward model, or inference-time state.

3.2 A GAP- AND SENSITIVITY-CONDITIONED FACTOR

We construct the structured factor offline from a calibration set. Let $q_t = q_0(\cdot | x, s, y_{<t})$ and let W be frozen. In the information-geometric view of local model sensitivity (Amari, 1998), the output Fisher pulled back to hidden space is

$$G_t = W^\top [\text{Diag}(q_t) - q_t q_t^\top] W, \quad G = \mathbb{E}_t[G_t]. \quad (6)$$

At each position, the implementation estimates G_t from 16 sampled LM-head rows rather than forming the full-vocabulary covariance. If $G = U \text{Diag}(\lambda) U^\top$, we apply the smooth spectral weight

$$a_\mu(\lambda) = \frac{\lambda}{\mu} \exp\left(1 - \frac{\lambda}{\mu}\right), \quad (7)$$

where μ is the geometric median of the positive eigenvalues. This weight suppresses directions that are nearly invisible to the output head and directions already carrying the largest local output sensitivity.

We also estimate the privileged residual covariance

$$C_\Delta = \mathbb{E}_t[e_t e_t^\top]. \quad (8)$$

In the Fisher eigenbasis, define

$$\widetilde{M} = \text{Diag}(\sqrt{a}) U^\top C_\Delta U \text{Diag}(\sqrt{a}). \quad (9)$$

If V_r contains the top r eigenvectors of $\widetilde{M}$, the structured factor is

$$F_{\text{struct}} = U \text{Diag}(\sqrt{a}) V_r. \quad (10)$$

We use $r = 64$. The factor is fitted once and remains frozen during training. We call this operator Structured Covariance and Output-Fisher Projected Error (SCOPE), and the resulting OPSD method SCOPE-OPSD.

Algorithm 1 SCOPE-OPSD calibration and training

Input: OPSD data $\mathcal{D} = \{(x, s)\}$, frozen base model q_0 , rank r , and target hidden-gradient RMS γ .

1. **Calibrate once.** On 128 calibration prompts, sample aligned positions from student trajectories; record e_t , q_t , and LM-head rows.
2. Estimate C_Δ and G ; apply equation (7); compute the top- r factor F_{struct} using equation (10).
3. Construct F_{rand} with the same nonzero spectrum but an independent orthonormal orientation.
4. For each factor F , measure $\text{RMS}(\nabla_\theta \mathcal{L}_{\text{hid}}(F))$ at initialization and set α_F to match γ .
5. **Train.** For each optimizer step, sample one student rollout, evaluate student and frozen privileged teacher on its aligned prefixes, compute equations (2) and (4), and update only the student LoRA with equation (5).

Output: one deployable student checkpoint; discard F at inference.

3.3 MATCHED ORIENTATION CONTROL

Our primary control replaces the orientation of F_{struct} while preserving its nonzero spectrum. In particular, write

$$F_{\text{struct}} F_{\text{struct}}^\top = P \text{Diag}(\sigma_1^2, \dots, \sigma_r^2) P^\top. \quad (11)$$

For an orthonormal frame $Q \in \mathbb{R}^{d \times r}$ satisfying $Q^\top Q = I_r$, the matched random factor is

$$F_{\text{rand}} = Q \text{Diag}(\sigma_1, \dots, \sigma_r). \quad (12)$$

It therefore has the same rank, trace, and nonzero spectrum as the structured factor, but its orientation is independent of the privileged residual. We sample Q independently of the privileged data from an orthonormal frame; this control construction supports arbitrary hidden dimension d .

Equal trace does not imply equal parameter updates because the two factors may align differently with the model Jacobian. At initialization, let

$$g_F = \text{RMS}(\nabla_\theta \mathcal{L}_{\text{hid}}(F)). \quad (13)$$

For a common target γ , we set

$$\alpha_F = \gamma / g_F. \quad (14)$$

Pure OPSD sets $\alpha_F = 0$. Structured and Random are alternative orientations of the same rank- r auxiliary, rather than additive components; their matched comparison is the main test of the proposed geometry.

3.4 LOCAL INTERPRETATION OF THE STRUCTURED FACTOR

The construction in equations (6) and (10) ranks directions using a local proxy: privileged discrepancy weighted by sensitivity through the output head. The offline diagnostic tests whether the fitted factor captures this target on held-out prompts. Downstream evaluation separately tests whether the local orientation improves autoregressive reasoning.

4 EXPERIMENTAL PROTOCOL

Compute environment. Every training run uses one 16-device Alibaba T-Head Zhenwu 810E (ZW810E) PPU node, with 96 GB HBM2e per device. The actor is sharded across all devices with FSDP. We use PyTorch 2.9.0 and the project VERL snapshot with its PPU-compatible backend; evaluation is served with SGLang. All matched arms at a given model scale use the same hardware class and software environment.

Training data and OPSD recipe. We use the 29,434-example OpenThoughts Math OPSD split (Guha et al., 2025; Zhao, 2026) and Qwen3-1.7B, 4B, and 8B (Yang et al., 2025). The student samples one non-thinking response per problem with temperature 1.1, top-p 0.95, top-k 20, and a 1,024-token limit. The privileged teacher is the frozen step-zero model in thinking mode; it receives the verified solution and the same student prefix. We follow the OPSD recipe: full-vocabulary forward KL, pointwise clip 0.05, LoRA rank 64 and alpha 128 (Hu et al., 2022), learning rate 5×10^{-6} , and effective batch size 32. Each arm runs for 100 optimizer steps and saves steps 25, 50, 75, and 100.

Scale-specific calibration. At each scale, we estimate G , C_Δ , and a rank-64 factor from 128 calibration prompts. Factors are neither padded nor transferred between models. The hidden widths are 2,048, 2,560, and 4,096. We also recalibrate the auxiliary coefficient at each scale using equation (14); the rank, training schedule, and evaluation protocol remain unchanged.

Compared systems. Base is the untrained model. Pure OPSD uses only equation (2). Matched Random uses equation (12), and Structured uses equation (10). Within a matched run, Pure, Random, and Structured use the same training split, batch and checkpoint schedules, rollout configuration, frozen teacher, adapter, optimizer, and output objective. Random and Structured receive the same initial hidden-gradient RMS. The comparison therefore adds no rollout samples, optimizer steps, or privileged inputs beyond Pure OPSD.

We additionally train a local Group Relative Policy Optimization (GRPO) reference under the source-paper recipe for 500 steps, saving checkpoints 100, 200, 300, 400, and 500 (Zhao et al., 2026a). GRPO changes both the objective and the schedule, so it is a standard post-training context row, not a control for the Pure–Random–Structured comparison.

Ablation variants. At Qwen3-1.7B and fixed step 75, GapOnly fits a rank-64 factor from privileged residual covariance without Fisher conditioning. FullHidden penalizes all $d = 2,048$ final-state coordinates. The rank sweep applies the complete Structured construction with $r \in \{32, 64, 128\}$. These variants retain the same OPSD output objective and evaluation protocol. They test component ordering and selective capacity; they are not used to retune the frozen cross-scale comparison.

Checkpoint policy. Based on the Qwen3-1.7B checkpoint trajectory, we use step 75 as the common checkpoint for the primary cross-scale comparison and apply the same choice to Qwen3-4B and Qwen3-8B. We additionally report all saved checkpoints at steps 25, 50, 75, and 100 to characterize convergence. Within each comparison, every benchmark uses the same checkpoint; we do not construct benchmark-wise checkpoint oracles.

This distinction matters for external context. The original OPSD paper reports benchmark-wise best values drawn from its checkpoint sweep, which need not correspond to one deployable model; it separately provides a single step-100 value for Qwen3-1.7B (Zhao et al., 2026a). PHF reports an independent step-100 OPSD reproduction (Li et al., 2026b). We label these selection rules when discussing external numbers and compute all method deltas only between our locally matched methods.

Table 1: Primary matched comparison at the common step-75 checkpoint under the OPSD Think Avg@12 protocol. All methods use the same training and evaluation recipe within each scale; deltas are percentage points.

Scale	Base	Pure OPSD	Matched Random	Structured	Struct.–Pure	Struct.–Random
Qwen3-1.7B	37.41	41.48	41.94	43.33	+1.85	+1.39
Qwen3-4B	60.65	62.13	63.43	63.80	+1.67	+0.37
Qwen3-8B	62.69	64.45	64.35	65.28	+0.83	+0.93

Table 2: Qwen3-1.7B Avg@12 at the common step-75 checkpoint.

Dataset	Matched random	Structured	Structured–Random
AIME24	57.78	58.33	+0.56
AIME25	40.70	42.50	+1.81
HMMT25	27.36	29.17	+1.81
Macro	41.94	43.33	+1.39

Evaluation. We follow the OPSD thinking-mode protocol on AIME 2024, AIME 2025, and HMMT 2025, with 30 problems per benchmark. Each problem receives 12 samples at temperature 1.0 and top-p 0.95, with at most 38,912 new tokens. If $c_{m,d,i,j} \in \{0, 1\}$ denotes correctness for method m , benchmark d , question i , and sample j , then

$$\hat{A}_{m,d} = \frac{100}{30 \cdot 12} \sum_{i=1}^{30} \sum_{j=1}^{12} c_{m,d,i,j}, \quad \hat{M}_m = \frac{1}{3} \sum_d \hat{A}_{m,d}. \quad (15)$$

Thus Avg@12 is generation-level accuracy, and the reported Macro is the unweighted mean across the three benchmarks.

Evaluation integrity. Every system is evaluated under the same sampling configuration. Each reported checkpoint covers all 90 problems with 12 samples per problem, yielding 1,080 generations. Only complete evaluations with zero failed requests enter the tables. Before aggregation, we verify the actor checkpoint, LoRA merge provenance, served-model identity, dataset hashes, decoding configuration, and generation count.

5 RESULTS

5.1 THE FIXED STEP-75 COMPARISON TRANSFERS ACROSS SCALES

Structured is the strongest local arm at all three scales in table 1. Its advantage over Pure OPSD ranges from 0.83 to 1.85 points, and it also exceeds the spectrum- and gradient-matched random orientation at every scale. The checkpoint is identical across the table and is shared by every benchmark and model scale. Even when Pure OPSD is allowed its best saved checkpoint in $\{25, 50, 75, 100\}$, fixed step-75 Structured remains higher by 1.85, 1.21, and 0.65 points at 1.7B, 4B, and 8B.

5.2 STRUCTURED ORIENTATION IMPROVES OVER MATCHED RANDOM AT 1.7B

At step 75, Structured improves over Matched Random by 0.56, 1.81, and 1.81 points on AIME24, AIME25, and HMMT25, respectively, yielding a +1.39-point Macro gain. The improvement is therefore not driven by a single benchmark. Across the complete checkpoint trajectory, Structured also maintains a higher Macro score at steps 50, 75, and 100 (table 7).

For context, the original OPSD study reports 43.40 for Qwen3-1.7B after taking the best checkpoint separately for each benchmark, and 42.50 for its single step-100 checkpoint (Zhao et al., 2026a). Our Structured model reaches 43.33 at the common step-75 checkpoint. At 4B and 8B, the corresponding

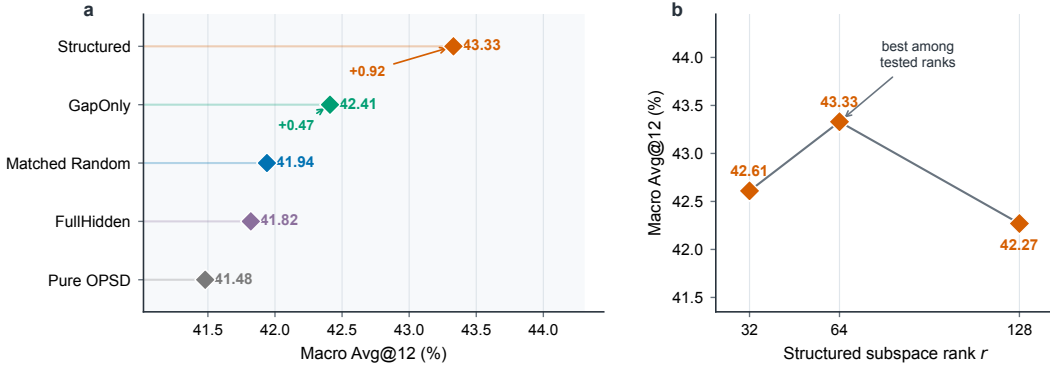


Figure 2: Qwen3-1.7B ablations at step 75 under the same Think Avg@12 protocol. **(a)** Component ordering and the FullHidden control. **(b)** Rank sensitivity of the complete Structured construction. Each marker denotes the reported method-level aggregate.

Structured scores are 63.80 and 65.28, compared with the source paper’s benchmark-wise checkpoint oracles of 63.60 and 64.80. Because the checkpoint-selection rules differ, these external values provide scale context only and are not used to compute the matched deltas in table 1.

5.3 A LOCAL GRPO CURVE ANCHORS THE POST-TRAINING CONTEXT

The locally trained Qwen3-1.7B GRPO curve obtains Macro Avg@12 values of 36.94, 38.24, 39.44, 38.80, and 38.61 at steps 100–500. Its best single deployable checkpoint is step 300 at 39.44, above the local Base score of 37.41 and below Pure OPSD at 41.48 and Structured at 43.33. Structured is 3.89 points higher than this local GRPO checkpoint. This is a contextual comparison rather than a matched causal delta: GRPO uses a different objective and a 500-step schedule, whereas the OPSD arms share a 100-step distillation recipe. The full curve and source-paper context appear in table 10.

5.4 COMPONENT AND RANK CONTROLS SUPPORT SELECTIVE STRUCTURE

Figure 2 separates two questions. First, the component sequence is Random < GapOnly < Structured: $41.94 < 42.41 < 43.33$. Matched Random is 0.46 points above Pure OPSD, GapOnly is 0.47 points above Random, and Structured is another 0.92 points above GapOnly. Under the tested setup, this ordering supports incremental contributions from a hidden auxiliary, privileged residual information, and Fisher conditioning. FullHidden averages 41.82, 0.12 points below Random and 1.51 below Structured; constraining more coordinates is therefore not monotonically beneficial in this setting.

Second, rank 64 is the strongest of the tested Structured ranks. It exceeds ranks 32 and 128 by 0.72 and 1.06 points. Together, these controls support the use of a privileged, Fisher-conditioned, selective low-rank subspace under the tested setup; they do not imply a unique causal decomposition or that rank 64 is universally optimal.

5.5 COMPLETE TRAJECTORIES AND THE TERMINAL CHECKPOINT

Structured is never below Pure OPSD in any of the 12 scale-checkpoint comparisons in figure 3: it is strictly above in 11 and exactly tied at 4B step 25 (657 versus 657 correct generations). It exceeds Random in 10 of 12; the exceptions are 1.7B step 25 and 4B step 50, each by 0.09 points. At the terminal step 100, the Structured–Random differences are +3.10, +0.93, and +1.94 points for 1.7B, 4B, and 8B. The corresponding Structured scores are 42.46, 63.43, and 66.11. Thus the primary step-75 result is not produced by choosing a different checkpoint for each benchmark, and the ordering persists at the shared budget endpoint.

The trajectories also show why the terminal checkpoint should not be treated as universally optimal. Structured peaks at step 75 for 1.7B and 4B, but at step 100 for 8B. Reporting the full curve separates this convergence behavior from the common cross-scale comparison.

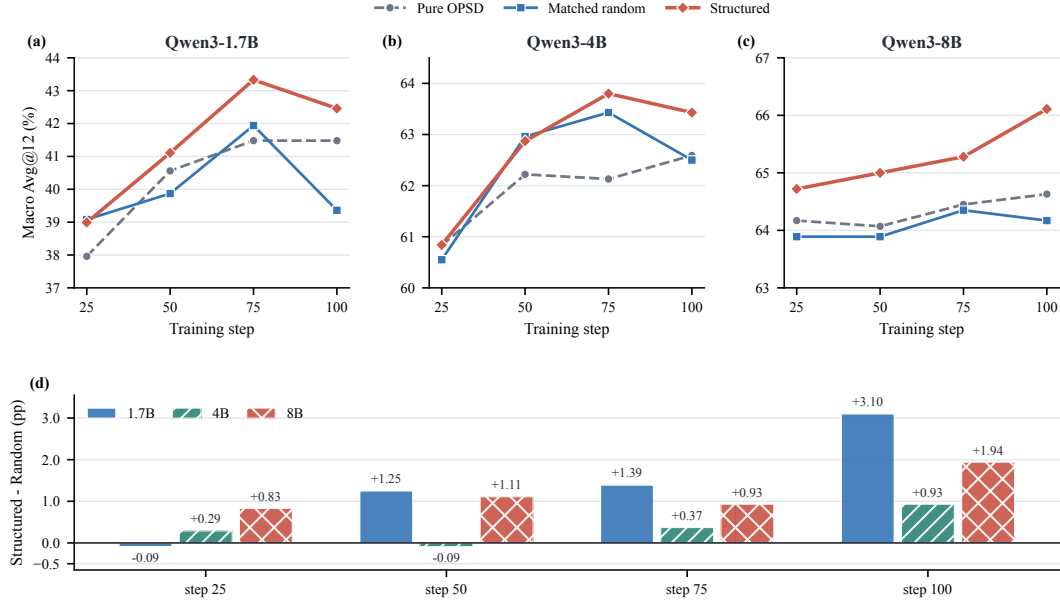


Figure 3: Complete checkpoint trajectories. (a–c) Macro Avg@12 at steps 25, 50, 75, and 100 for Pure OPSD, spectrum- and gradient-matched Random, and Structured. (d) Structured minus Random for each matched scale-checkpoint pair. All points use the same method-level reporting protocol.

5.6 THE FITTED FACTOR CAPTURES ITS OFFLINE TARGET

On held-out calibration halves, the structured factor captures $4.40\times$ the privileged-gap score of a spectrum-matched random orientation. Under a local full-vocabulary intervention, its mean KL reduction is 0.2347, compared with 0.0241 for Random. These diagnostics verify that the factor is data dependent and selects the intended local geometry; they do not replace the downstream test, which measures correctness after autoregressive rollout. Section E gives the estimands and complete diagnostic table.

6 RELATED WORK

Knowledge and representation distillation. Classical distillation matches teacher and student distributions (Hinton et al., 2015). Representation variants transfer intermediate features, attention, relations, or contrastive structure (Romero et al., 2015; Zagoruyko & Komodakis, 2017; Tian et al., 2020), including hidden layers in Transformers (Sun et al., 2019; Wang et al., 2020). Unlike these fixed-distribution settings, we compare ordinary and solution-conditioned views of one model along its own trajectories.

On-policy language-model distillation. Generalized Knowledge Distillation trains on student sequences (Agarwal et al., 2024); later work adapts divergences or token weights (Ko et al., 2025; Zhang et al., 2025; Jung et al., 2025; Song & Zheng, 2026). OPSD instead gives a frozen self-teacher a verified solution and retains dense full-vocabulary supervision on student prefixes (Zhao et al., 2026a). Subsequent studies examine teacher bias, token dynamics, and supervision scope (Li et al., 2026a; Jiang et al., 2026b; Harne et al., 2026; Zhao et al., 2026b; Hou et al., 2026; Yang et al., 2026b). OP2SD changes the privileged example (Ichihara et al., 2026), PAST conditions on the completed student trajectory and outcome (Feng et al., 2026), and SA-OPD filters weakly grounded signals (Jiang et al., 2026a). Recent SCOPE calibrates OPD by routing correct and incorrect rollouts through separate, perplexity-weighted supervision paths (Zheng et al., 2026). By contrast, SCOPE-OPSD leaves rollout routing and the full-vocabulary OPSD objective unchanged, adding an auxiliary loss on a frozen Fisher-conditioned privileged-residual subspace.

Hidden-space and geometric supervision. OPRD aligns selected hidden representations and can replace output-space distillation (Yang et al., 2026a). The contemporaneous PHF preprint augments OPSD with hidden transitions, within-layer trajectory geometry, and neighboring-layer relations (Li et al., 2026b). Its ablations remove additive loss components; we instead test one final-layer privileged-residual orientation against a rank-, spectrum-, and gradient-matched replacement. Latent On-Policy Self-Distillation (LOPD) makes the teacher’s privileged context learnable end to end as continuous latent tokens, while retaining dense token-level supervision at each visited prefix (Zhang et al., 2026). FP-OPD uses Fisher/tangent geometry for an output correction in vision-language distillation (Xue et al., 2026); we use output Fisher only to construct a frozen hidden factor for language-model self-distillation. Related analyses motivate separating local update geometry from sequence-level correctness (Shen et al., 2026; Ge et al., 2026).

7 LIMITATIONS

Our experiments cover one model family and three competition-mathematics benchmarks. Other architectures, reasoning domains, and longer training budgets may exhibit different convergence behavior. Avg@12 measures accuracy under a fixed sampling budget rather than large- K solvability.

The matched-random comparison controls rank, spectrum, and initial hidden-gradient scale. The GapOnly ordering separates residual covariance from Fisher conditioning, while FullHidden and the rank sweep support a selective low-rank design. These mechanism controls are currently limited to 1.7B. Their ordering supports the roles of privileged residual information, Fisher conditioning, and selective rank under the tested setup, but does not establish a unique causal decomposition or a rank that is optimal beyond the tested set.

8 CONCLUSION

SCOPE-OPSD adds a frozen Fisher-conditioned privileged subspace to the existing OPSD computation. It reuses aligned student and teacher states, requires no extra rollout, and leaves deployment unchanged. At the common step-75 checkpoint, Structured improves both Pure OPSD and a spectrum- and gradient-matched random orientation on Qwen3-1.7B, 4B, and 8B. The complete checkpoint curves preserve the broader ordering through the shared 100-step budget and avoid benchmark-specific checkpoint selection. The cross-fitted diagnostic connects this downstream behavior to the intended privileged-gap geometry. Component controls place GapOnly between Random and Structured, while full-state matching and the tested ranks 32 and 128 remain below rank 64. A locally trained GRPO curve provides a standard post-training reference without being treated as a matched objective or budget comparison. Together, these results support a compact hidden channel for short-budget on-policy self-distillation.

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A REPRODUCIBILITY DETAILS

A.1 TRAINING CONFIGURATION

Table 3: Shared training configuration.

Setting	Value
Compute node	16× Alibaba T-Head Zhenwu 810E (ZW810E) PPUs
Device memory	96 GB HBM2e per PPU
Training stack	PyTorch 2.9.0; VERL with PPU-compatible backend; FSDP
Evaluation serving	SGLang
Training split	OpenThoughts Math OPSD, 29,434 examples
Optimizer / saved steps	100 / {25, 50, 75, 100}
Effective batch size	32
Student rollout	non-thinking, one response, at most 1,024 tokens
Teacher	thinking, frozen step-zero self-teacher
Rollout sampling	temperature 1.1, top-p 0.95, top-k 20
Output objective	full-vocabulary forward KL; clip 0.05
Adapter	LoRA rank 64, alpha 128
Learning rate	5×10^{-6}
Hidden rank / calibration prompts	64 / 128

Table 4: Scale-specific factor calibration.

Scale	Hidden / layers	α_{rand}	α_{struct}
1.7B	2,048 / 28	6.3589×10^{-4}	2.5626×10^{-4}
4B	2,560 / 36	6.7885×10^{-4}	2.4665×10^{-4}
8B	4,096 / 36	8.3017×10^{-4}	3.6154×10^{-4}

All three models use vocabulary size 151,936, auxiliary rank 64, and a 100-step training budget.

The matched random control samples an orthonormal frame independently of the privileged data. The implementation uses randomized normalized Hadamard columns when d is a power of two and reduced QR of a pseudorandom Gaussian matrix otherwise. Both constructions satisfy $Q^\top Q = I_r$ and affect only the counterfactual control in equation (12); the structured method is unchanged across hidden widths.

A.2 EVALUATION AND STATISTICAL ESTIMANDS

Table 5: Evaluation configuration.

Setting	Value
Benchmarks	AIME24, AIME25, HMMT25
Problems / samples per problem	30 / 12
Maximum new tokens	38,912
Sampling	temperature 1.0, top-p 0.95
Prompt mode	Qwen3 thinking mode
Metric	Avg@12; three-benchmark macro
Generations per evaluation	1,080

Equation 15 defines the point estimators. For a matched pair of methods a and b , we report $\hat{\Delta}_{a,b} = \hat{M}_a - \hat{M}_b$. Tables and figures report method-level aggregates under the common evaluation protocol.

Every result row has 1,080/1,080 generations and zero failed requests. We also verify source actor step, LoRA merge provenance, served-model identity, dataset hashes, and decoding configuration before result aggregation.

A.3 ANALYTICAL INCREMENTAL COST

Table 6: Incremental online cost relative to Pure OPSD. T is response length, d hidden width, and $r = 64$.

Quantity	Increment
Student rollouts / teacher forwards	0 / 0
Projection arithmetic	$\Theta(Tdr)$
Projected activations / frozen factor	$O(Tr) / dr$ scalars
Trainable parameters	0
Inference parameters / FLOPs	0 / 0

The factor contains 131,072, 163,840, and 262,144 scalars at 1.7B, 4B, and 8B. Offline calibration uses 128 prompts, stores dense $d \times d$ moment matrices, and performs one dense spectral solve per factor fit. This one-time stage is not part of online training and the factor is discarded after training. The table reports analytical properties of the added interface rather than end-to-end system measurements.

A.4 DATA PROVENANCE

We use the released OPSD training split without adding AIME24, AIME25, or HMMT25 examples. Evaluation files are immutable 30-problem sets with recorded hashes, preserving the source recipe’s train/evaluation boundary.

B COMPLETE CHECKPOINT CURVES

B.1 QWEN3-1.7B

Table 7: Macro Avg@12 across the complete Qwen3-1.7B checkpoint trajectory.

Step	Pure	Random	Structured	Struct.–Pure	Struct.–Random
25	37.96	39.08	38.99	+1.03	-0.09
50	40.56	39.87	41.11	+0.55	+1.25
75	41.48	41.94	43.33	+1.85	+1.39
100	41.48	39.36	42.46	+0.98	+3.10

B.2 QWEN3-4B

Table 8: Complete Qwen3-4B Avg@12 trajectory.

Arm	Step	AIME24	AIME25	HMMT25	Macro
Base	0	73.61	65.56	42.78	60.65
Pure	25	73.61	66.11	42.78	60.83
Pure	50	74.72	67.50	44.44	62.22
Pure	75	75.56	67.78	43.06	62.13
Pure	100	76.11	67.22	44.44	62.59
Random	25	74.44	66.39	40.83	60.55
Random	50	76.67	68.33	43.89	62.96
Random	75	76.11	67.78	46.39	63.43
Random	100	75.56	68.06	43.89	62.50
Structured	25	74.17	67.78	40.56	60.84
Structured	50	74.44	70.83	43.33	62.87
Structured	75	75.28	69.44	46.67	63.80
Structured	100	73.06	70.83	46.39	63.43

B.3 QWEN3-8B

Table 9: Complete Qwen3-8B Avg@12 trajectory.

Arm	Step	AIME24	AIME25	HMMT25	Macro
Base	0	76.67	65.56	45.83	62.69
Pure	25	76.67	71.11	44.72	64.17
Pure	50	75.83	70.83	45.56	64.07
Pure	75	75.28	72.50	45.56	64.45
Pure	100	77.22	70.28	46.39	64.63
Random	25	77.22	68.33	46.11	63.89
Random	50	78.06	68.33	45.28	63.89
Random	75	77.50	69.72	45.83	64.35
Random	100	77.50	69.17	45.83	64.17
Structured	25	76.11	72.78	45.28	64.72
Structured	50	78.06	72.78	44.17	65.00
Structured	75	78.61	71.39	45.83	65.28
Structured	100	76.94	73.89	47.50	66.11

C LOCAL GRPO POST-TRAINING CONTEXT

Table 10: Complete local Qwen3-1.7B GRPO checkpoint curve under the official Think Avg@12 protocol.

Step	AIME24	AIME25	HMMT25	Macro
100	51.11	36.94	22.78	36.94
200	52.50	38.61	23.61	38.24
300	56.11	38.33	23.89	39.44
400	52.78	38.06	25.56	38.80
500	52.22	38.33	25.28	38.61

The source paper reports a GRPO peak within 500 steps of 51.10/38.30/23.70, or 37.70 Macro (Zhao et al., 2026a). Our local best single checkpoint is step 300 at 39.44. Selecting the best local

checkpoint separately for each benchmark would produce 56.11/38.61/25.56, or 40.09 Macro, but this is a non-deployable diagnostic oracle and is not used as a method result. GRPO and OPSD use different objectives and 500- versus 100-step schedules; the curve is therefore kept separate from the matched Pure–Random–Structured comparison.

D COMPONENT AND RANK ABLATIONS

Table 11: Qwen3-1.7B component and rank ablations at step 75 under the common Think Avg@12 protocol.

Variant	Rank	AIME24	AIME25	HMMT25	Macro
Pure OPSD	–	54.17	43.89	26.39	41.48
Matched Random	64	57.78	40.70	27.36	41.94
GapOnly	64	56.39	41.11	29.72	42.41
FullHidden	$d = 2,048$	56.94	40.74	27.78	41.82
Structured	32	56.53	42.13	29.17	42.61
Structured	64	58.33	42.50	29.17	43.33
Structured	128	57.22	40.70	28.89	42.27

FullHidden is +0.34 points above Pure but -0.12 below Random; Structured is +1.51 above FullHidden. GapOnly is +0.47 above Random and -0.92 below Structured. Finally, rank 64 is +0.72 and +1.06 above ranks 32 and 128, respectively. The ordering supports the structured selective-subspace design within the tested setting.

E OFFLINE FACTOR DIAGNOSTICS

The Qwen3-1.7B audit uses 128 prompts and 4,096 sampled response positions. It partitions prompts into two complementary folds, fits a factor on one fold, and evaluates it only on the other. For held-out positions $\mathcal{H}$, define the normalized privileged-gap score

$$S_{\text{gap}}(F; \mathcal{H}) = \frac{1}{|\mathcal{H}|} \sum_{t \in \mathcal{H}} \frac{\|F^\top e_t\|_2^2}{\text{tr}(F^\top F)}. \quad (16)$$

This is the cross-fitted quantity behind the reported $4.40\times$ ratio.

To measure overlap with the existing output objective, let $g_t = \nabla_{h_t^S} D_{\text{KL}}(q_t \| p_t)$. The output-gradient score is

$$S_{\text{grad}}(F) = \mathbb{E}_t \left[\frac{\|F^\top g_t\|_2^2}{\text{tr}(F^\top F)} \right]. \quad (17)$$

Finally, with intervention strength $\eta = 0.5$, we move the student state along the projected residual,

$$h_t^{S,F} = h_t^S - \eta F F^\top e_t, \quad (18)$$

and define the exact local KL gain

$$\Delta_{\text{KL}}(F) = \mathbb{E}_t \left[D_{\text{KL}}(q_t \| p(\cdot | h_t^S)) - D_{\text{KL}}(q_t \| p(\cdot | h_t^{S,F})) \right]. \quad (19)$$

The positive-intervention ratio is the fraction of positions whose bracketed difference in equation (19) is positive.

Table 12: Frozen Qwen3-1.7B offline diagnostics.

Diagnostic	Structured	Random	Ratio / delta
A fit $\rightarrow$ B gap	344.36	79.35	$4.34\times$
B fit $\rightarrow$ A gap	335.30	75.25	$4.46\times$
Full-data gap / trace	9.02	1.37	$6.59\times$
Output-gradient / trace	3.86×10^{-5}	7.43×10^{-5}	-48.1%
Exact KL gain	0.2347	0.0241	$9.75\times$
Positive intervention	60.55%	60.30%	+0.25 pp

The two cross-fitted rank-64 bases have canonical cosine 0.369, indicating stable held-out score without identical coordinates. Structured intervention gain is 0.6544, 0.0023, and 0.0264 in early, middle, and late response buckets. The 4B and 8B pipelines repeat the same integrity and cross-fit checks before training; none of these diagnostics is used as a substitute for downstream evaluation.

REPRODUCIBILITY STATEMENT

Section 4 specifies the training, checkpoint, and evaluation protocols. Appendix A reports hyperparameters, hardware, statistical estimands, analytical incremental costs, complete checkpoint curves, and the offline calibration audit. The accompanying arXiv source bundle contains the manuscript sources, bibliography, and final figures.